

CUSTOMER FEATURE GUIDE

Zyvor Fabric

A systemd-native private cloud control plane.

Run enterprise-grade virtual machines, networking, storage, and security on any Linux server — with one binary, one config, and one systemd service.

by ZyvorAI Labs

What is Zyvor Fabric?

Zyvor Fabric wraps systemd-vmspawn and systemd-machined in a complete Rust control plane, giving you Proxmox- and KubeVirt-class capabilities without the heavyweight stack. Manage the same infrastructure five ways — CLI, terminal UI, web dashboard, Kubernetes operator, or Terraform — over a single daemon exposing 480+ REST endpoints and live WebSocket channels. VMs become first-class systemd units with journal logging, watchdogs, and socket activation, so there are no custom hypervisor patches or kernel modules to maintain.

480+

REST API endpoints

5

management interfaces

40+

Rust backend crates

6

storage backends

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VM Lifecycle & Provisioning

Create, run, and reshape virtual machines on systemd-vmspawn with declarative or interactive workflows.



Full VM Lifecycle

Create, start, stop, restart, pause, resume, hibernate, and delete VMs backed by systemd-vmspawn and KVM.

[One consistent lifecycle across CLI, TUI, web, Terraform, and Kubernetes.](#)



Declarative Apply

Define VMs in YAML and reconcile them with vmctl apply -f config.yaml.

[GitOps-friendly infrastructure without a control-plane rewrite.](#)



Cloning & Templates

Full and linked copy-on-write clones plus reusable templates for rapid deployment.

[Stand up fleets from a golden image in seconds, not minutes.](#)



Hibernate & Checkpoint

Suspend-to-disk hibernate, resume from snapshot, and VM checkpoint/restore and forking.

[Pause idle workloads and restore exact machine state on demand.](#)



Live Hotplug

Hotplug CPU, memory, disk, and NIC into running VMs without a reboot.

[Scale a VM to demand while it keeps serving traffic.](#)



Disk Import & Conversion

Import VMs from VMDK, VDI, and VHD with auto-conversion to qcow2, and online disk resize via QMP.

[Bring machines off VMware or VirtualBox without downtime.](#)

VMs are ordinary systemd units — journalctl, watchdogs, and socket activation work exactly as operators already expect.

Storage Orchestration

Six pluggable backends, live disk mobility, and a built-in cloud-image catalog.



Six Storage Backends

Pool and volume management across Local, NFS, LVM, LVM-thin, ZFS, and Ceph/RBD.

[Use the storage you already run — no dedicated SAN required.](#)



Volume Management

Full volume CRUD with attach/detach, online resize, and clone operations.

[Reshape storage for a workload without recreating the VM.](#)



Snapshots & Retention

Create and restore snapshots with configurable retention policies.

[Roll back a bad change in seconds and prune old state automatically.](#)



Storage Live Migration

Move VM disks between pools with no downtime, guided by SDRS recommendations.

[Rebalance or evacuate storage while VMs stay online.](#)



ZFS Replication

Incremental ZFS send/receive replication between hosts and sites.

[Efficient, block-level DR copies that only ship the delta.](#)



Cloud Image & ISO Catalog

Built-in downloader for Ubuntu, Fedora, Debian, and Alma images plus ISO download/list/delete.

[Boot a fresh distro without hunting for images.](#)

Software-Defined Networking

A full SDN stack — policies, firewalling, load balancing, VPN mesh, and observability.



Network Policies

Cilium-style label-based ingress/egress rules enforced through nftables.

Segment east-west traffic with identity, not brittle IP lists.



Per-VM Firewall

Firewall profiles and zones applied per VM via nftables, with IPv6 dual-stack support.

Ship each workload with its own hardened perimeter.



Service Mesh

Virtual-IP load balancing with round-robin, least-connection, random, and IP-hash strategies plus health checks.

Front a pool of VMs behind one resilient endpoint.



WireGuard VPN Mesh

Point-to-point, hub-spoke, and full-mesh WireGuard overlay tunnels.

Securely stitch VMs across sites without external appliances.



QoS Traffic Shaping

Guaranteed and maximum rates, burst, and priority queuing via Linux tc.

Protect critical workloads from noisy-neighbor bandwidth spikes.



DNS Policy & NAT Gateway

Zone management, upstream servers, domain blocking, plus SNAT/DNAT/hairpin NAT gateways.

Own DNS and egress routing for every tenant network.



Packet Mirror & Net Monitor

Mirror sessions for traffic capture and per-VM bandwidth tracking with threshold alerts.

Debug and meter network behavior without leaving the fabric.

Security & Identity

JWT auth, enterprise SSO, RBAC, multi-tenancy, and encryption on every endpoint.



JWT Auth + RBAC

JWT authentication with configurable expiry and 3-tier RBAC (Admin/User/Viewer) enforced on every endpoint.

Least-privilege access is the default, not an add-on.



Enterprise SSO

LDAP and OIDC/OAuth2 integration plus API keys for service-to-service auth.

Plug into existing identity providers instead of a new user silo.



Multi-Tenancy

Project isolation with member roles and per-project quotas.

Give teams their own bounded slice of the fabric.



Encryption & Secrets

Encryption at rest, per-VM disk encryption with key rotation, and an encrypted secrets store with access policies.

Protect data and credentials without external KMS scaffolding.



Audit Logging

Audit trail on all VM lifecycle operations with JSON/CSV export.

Answer 'who did what, when' for compliance and RCA.



PKI & Certificate Manager

CA creation, certificate issue/renew/revoke, automated rotation, and hardware attestation.

Run internal TLS without a separate PKI product.



Compliance Scanning

Built-in CIS, STIG, and PCI-DSS profiles with per-VM findings and remediation guidance.

Prove and improve posture against recognized benchmarks.

The codebase carries a documented multi-round security audit: zero unsafe Rust, no shell pipelines, parameterized queries, SSRF and path-traversal protection, and bcrypt-hashed credentials.

High Availability & Disaster Recovery

Clustering, live migration, fault tolerance, and multi-site recovery.



etcd Clustering

Multi-node clustering on an etcd state store with leader election and heartbeat-based health.

No single control-plane node to take the fleet down with it.



Live Migration

Move running VMs between hosts with iterative rsync pre-copy and cutover, with progress tracking and cancel.

Drain a host for maintenance without stopping workloads.



Fault Tolerance & Fencing

Continuous VM replication with automatic failover detection, fencing, and FT metrics.

Survive a node loss with minimal recovery time.



Predictive DRS

Distributed resource scheduling with demand forecasting, proactive placement, and affinity/anti-affinity rules.

Keep clusters balanced before hotspots become outages.



Site Recovery

Recovery plans with planned migration, disaster failover, test failover, and reprotection workflows.

Rehearse and execute cross-site DR with confidence.



Multi-Site Replication

Cross-site VM replication with sync scheduling, RPO monitoring, and recovery instances.

Meet recovery-point targets you can actually measure.

Compute, GPU & Virtualization

Low-level control over CPU topology, memory, accelerators, and firmware.



GPU Passthrough

First-class GPU passthrough for NVIDIA, AMD, and Intel GVT-g on Linux KVM.

[Run AI, rendering, and CAD workloads at near-bare-metal speed.](#)



CPU Pinning & NUMA

CPU topology control, pinning, NUMA-aware placement, and nested virtualization.

[Squeeze predictable performance from latency-sensitive VMs.](#)



Memory Optimization

Memory ballooning, hugepages, and KSM page deduplication via a system resource manager.

[Fit more VMs per host without starving any of them.](#)



vTPM & Secure Boot

TPM 1.2/2.0 via swtpm with per-VM isolated state, UEFI, and Secure Boot firmware management.

[Support BitLocker, LUKS, and measured boot inside guests.](#)



cloud-init Provisioning

NoCloud datasource generation for users, packages, network config, and SSH key injection.

[Boot fully configured VMs on first start, hands-free.](#)



OVA/OVF & Content Library

OVA/OVF export and import plus a content library with cross-site image/template sync and customization specs.

[Standardize and share golden images across every site.](#)

Monitoring, Automation & Operations

Metrics, scheduling, notifications, and self-checks that keep the fabric healthy.



Prometheus Metrics

A /metrics endpoint exposing per-VM CPU, memory, disk, and network stats plus API latency, with a prebuilt Grafana dashboard.

[Drop into your existing observability stack instantly.](#)



Scheduling & Auto Backups

Once/daily/weekly VM schedules plus automated daily backups and weekly state-store cleanup via systemd timers.

[Routine operations run themselves, on time, every time.](#)



Backup & Restore

Per-VM and bulk backups with retention policies and incremental backups from web UI and TUI.

[Recover a single VM or the whole fleet on your own terms.](#)



Multi-Channel Notifications

Email, Slack, Microsoft Teams, and webhook alerts with retry and backoff.

[The right people hear about problems the moment they happen.](#)



Health & Auto-Verify

Deep health checks (API, disk, DB, timers, KVM) and post-install smoke tests of API, auth, VM CRUD, and backups.

[Catch a broken deploy before your users do.](#)



Config Snapshots & Events

Versioned config-snapshot API, retained lifecycle events, and an SSE event stream for time-machine correlation.

[Diff infrastructure over time and reconstruct incidents.](#)

Interfaces & Automation Surfaces

Five first-class ways to drive the same daemon — pick per task, not per product.

Interface	Best for	Highlights
vmctl CLI	Scripting & automation	JSON/YAML/table output, apply -f
vmctl-tui	Live terminal ops	vim keys, sparklines, per-VM actions
Web dashboard	Operators & teams	37+ pages, Ctrl+K palette, bulk ops
K8s operator	GitOps / K8s shops	VirtualMachine CRD reconciliation
Terraform / SDK	Infra-as-code	plan/apply, typed Rust + Python + Ansible



vmctl CLI

Scriptable command-line client with JSON/YAML/table output covering VM, policy, storage, and network operations.

[Automate anything the platform can do from a shell script.](#)



vmctl-tui

k9s-style ratatui terminal dashboard with vim keybindings, sparklines, and live per-VM actions.

[Fleet control from a terminal, no browser required.](#)



Web Dashboard

React web UI with 37+ pages, a Ctrl+K command palette, dark theme, live WebSocket updates, and bulk operations.

[Give operators a full GUI without giving up the API.](#)



Console & VNC

Browser terminal via xterm.js over WebSocket and graphical VNC via a noVNC proxy, authenticated with the same JWT.

[Reach any VM's console without exposing raw ports.](#)



Kubernetes Operator

Manage VMs as VirtualMachine CRDs with continuous reconciliation via a Helm-installable operator.

[Define VMs alongside containers in the same GitOps flow.](#)



Terraform, SDK & Ansible

A Terraform provider with full plan/apply, a typed Rust vmspawn-sdk, plus Python and Ansible SDKs.

[Provision the fabric from whatever IaC tooling your team already uses.](#)

Fleet, Cost & Governance

Datacenter hierarchy, resource pools, chargeback, and lifecycle compliance at scale.



Datacenter Hierarchy

Model datacenters, clusters, and hosts with registration, heartbeat, maintenance mode, and auto-discovery.

[Organize sprawling infrastructure into a navigable topology.](#)



Resource Pools & Quotas

CPU/memory/storage reservations with admission control, overcommit ratios, and pool-level quotas.

[Guarantee capacity to teams while preventing runaway usage.](#)



Billing & Chargeback

Per-VM metering, configurable pricing tiers, invoice generation, and chargeback reports.

[Show every team the true cost of what they run.](#)



Lifecycle Manager

Host baseline definitions, compliance scanning, remediation, and rolling updates with pause/advance.

[Keep every host on a known-good, patched baseline.](#)



Distributed Storage Policies

Datastore clusters, storage policies, SDRS recommendations, and compliance checking.

[Enforce storage placement rules automatically across pools.](#)

Deployment scales from a single 4GB edge server to a 3+ node etcd cluster with shared storage and HA failover — same binary, same API.

GETTING STARTED

Your first 30 minutes

A suggested path to value on day one.

- 1 Install in one command**
Clone the repo and run `make build && sudo make install`, or `./vmspawnctl deploy` for an auto-sudo end-to-end setup.
- 2 Start the service**
Run `sudo systemctl enable --now zyvor-fabric`, then read the auto-generated admin password with `./vmspawnctl password`.
- 3 Create your first VM**
Use `vmctl vm create --name web-01 --cpus 2 --memory 4G`, or declare it in YAML and run `vmctl apply -f config.yaml`.
- 4 Open your interface of choice**
Launch `vmctl-tui` in a terminal or open the web dashboard at <https://localhost:8443> to manage the fleet.
- 5 Verify and monitor**
Run `./vmspawnctl verify` and `./vmspawnctl health`, then scrape `/metrics` into Prometheus and import the bundled Grafana dashboard.

Support & contact

Zyvor Fabric is developed by **ZyvorAI Labs**. For onboarding, a proof-of-concept, or a guided demo, contact your account team at info@zyvor.dev.

Good to know: Zyvor Fabric requires Linux with systemd 256+ (Fedora, Ubuntu, Debian, RHEL, or SUSE) and KVM; it is not a hosted or Windows-server product. Some enterprise capabilities carry environmental prerequisites — swtpm for vTPM, an etcd cluster and shared/replicated storage for HA and live migration, and matching hardware/IOMMU for GPU passthrough. Multi-node HA is designed for 3+ nodes; single-server deployments run standalone. The published endpoint and page counts (480+ REST endpoints, 37+ web pages) reflect current documentation and may vary by release, and the macOS Machina workbench is a separate desktop product that consumes the Fabric API rather than part of this daemon.

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